

PCA and LDA

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PCA_
LDA_2405...

⇒ PCA is used for dimensionality reduction, lossy data compression, feature extraction, & data visualization.

* Maximum Variance Formulation:-

→ Want to project data onto a space having dimensions $M < D$ while maximizing variance of the projected data.

◦ Derivation :-

⇒ Consider that you want to project your data onto a one-dimensional space ($M=1$).

↳ Let a dimensional unit vector u represent this space (M)
i.e. $u^T u = 1$. $\therefore (u_1, u_2, \dots)$

⇒ Each data point x_n is then projected onto a scalar of value $u_1^T x_n$. The mean of the projected data is given by.

$$\mu_{\text{projected}} = \frac{1}{N} \sum_{n=1}^N u_1^T x_n.$$

$$\therefore \mu_{\text{projected}} = u_1^T \bar{x}, \text{ where } \bar{x} = \frac{1}{N} \sum_{n=1}^N x_n.$$

⇒ The variance is given as.

$$\begin{aligned} & \frac{1}{N} \sum_{n=1}^N (\text{value} - \text{mean})^2 \\ &= \frac{1}{N} \sum_{n=1}^N (u_1^T x_n - u_1^T \bar{x})^2. \end{aligned}$$

$$\text{Var}_{\text{projected}} = u_1^T S u_1.$$

⇒ where S is the data covariance matrix:-

$$S = \frac{1}{N} \sum_{n=1}^N (x_n - \bar{x})(x_n - \bar{x})^T.$$

✳ we maximize $u_1^T S u_1$ wrt u_1 due to prevent $\|u_1\| \rightarrow \infty$, we do $\|u_1\|=1$, we enforce this using Lagrangian:

$$L = \max_{u_1} u_1^T S u_1 + \lambda (1 - u_1^T u_1)$$

$$\therefore \frac{dL}{du} = 0 \Rightarrow 2 S u_1 = 2 \lambda u_1$$

$$\therefore S u_1 = \lambda u_1$$

$\therefore u_1$ is maximized when it is the eigen vector,

$$\therefore u_1^T S u_1 = \lambda_1 \quad (\because u_1^T u_1 = 1)$$

1. Form the data matrix

$$X = \begin{pmatrix} (x^{(1)})^T \\ (x^{(2)})^T \\ \vdots \end{pmatrix} \quad (\text{shape } n \times d).$$

→ Variance will be maximum when we set u_1 equal to the eigen vector having the largest eigen value λ .

∴ Optimal linear projection for which the variance of the projected data is maximized is now defined by the M eigen vectors $u_1, \dots, u_M$ of the data covariance matrix S corresponding to M largest eigen values.

- 1) evaluate mean $\bar{y}$ of dataset.
- 2) evaluate covariance S
- 3) find M eigen vectors of S with largest eigen values.
↳ Can be done using eigen decomposition.

$$(x^{(n)})^T$$

2. Compute the mean vector

$$\mu = \frac{1}{n} \sum_{i=1}^n x^{(i)} \in \mathbb{R}^d.$$

3. Center the data

$$\tilde{X} = X - \mathbf{1} \mu^T \quad (\text{subtract } \mu \text{ from each row}),$$

where $\mathbf{1} \in \mathbb{R}^n$ is a column of ones.

4. Form the sample covariance matrix

$$S = \frac{1}{n-1} \tilde{X}^T \tilde{X} \in \mathbb{R}^{d \times d}.$$

Equivalently, the (j, k) -entry is

$$S_{jk} = \frac{1}{n-1} \sum_{i=1}^n (x_j^{(i)} - \mu_j) (x_k^{(i)} - \mu_k).$$

* LDA (Dimensionality Reduction):-

* Shortcomings:-

- Linear decision boundary is lesser than the
- When the number of observations ~~is~~ the number of features, LDA might not perform as desired. This is called Small sample size (SSS) problem. Regularization is required.

* Assumptions:-

- Assumes data to be distributed normally, or gaussian distribution of data points.
↳ Each feature must make a bell-shaped curve when plotted.
- Each of the classes have identical covariance matrices.
- ∴ LDA performs quite well even if the assumptions are violated.

* Fischer's Linear Discriminant:-

- LDA is a general form of Fischer's Linear Discriminant.
- ∴ Basic idea of FLD is to project data points onto a line to maximize the between-class scatter & minimize the within class scatter.

→ The mean vector of each class in x & y feature space is

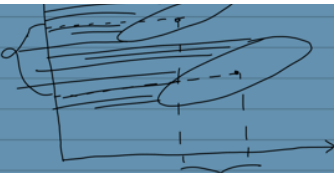
$$\mu_1 = \frac{1}{N_1} \sum_{x \in \omega_1} x \quad \text{and} \quad \tilde{\mu}_1 = \frac{1}{N_1} \sum_{y \in \omega_1} y = \frac{1}{N_1} \sum_{x \in \omega_1} w^T x = w^T \mu_1$$

→ where y is the projected feature space
i.e. $y = w^T x$.

→ The distance between projected means (means of classes after projection) is not a very good measure > since



The axis yield better class separability



this axis has a larger distance between means, between class separability is low.

- The solution proposed by Fisher is to maximize a function that represents the difference between means, normalized by a measure of within class scatter.

⇒ Scatter is defined as

$$\tilde{S}_1^2 = \sum_{y \in \omega_1} (y - \tilde{\mu}_1)^2 \Rightarrow \text{Equivalent of Variance.}$$

⇒ So total "scatter" or within class scatter (variance) is

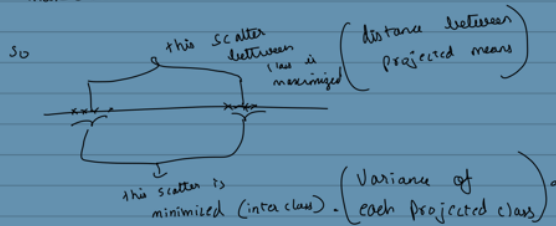
$$\tilde{S}_1^2 + \tilde{S}_2^2.$$

⇒ Fisher's linear discriminant is defined as the linear function $w^T x$ that maximizes the criterion function.

$$J(w) = \frac{|\tilde{\mu}_1 - \tilde{\mu}_2|^2}{\tilde{S}_1^2 + \tilde{S}_2^2} \rightarrow \text{Maximize.}$$

Minimize

max $J(w)$



$$\therefore S_i = \sum (x - \mu_i)(x - \mu_i)^T$$

$$S_1 + S_2 = S_w \quad (S_w \Rightarrow \text{within class scatter matrix}).$$

$$\begin{aligned} \therefore \tilde{S}_1^2 &= \sum_{y \in \omega_1} (y - \mu_1)^2 = \sum_{x \in \omega_1} (w^T x - w^T \mu_1)^2 \\ &= \sum_{x \in \omega_1} w^T (x - \mu_1)(x - \mu_1)^T w = w^T S_1 w \end{aligned}$$

$$\therefore \tilde{S}_1^2 + \tilde{S}_2^2 = w^T S_w w.$$

- The difference between projected means can be expressed in terms of means in the original feature space.

$$(\tilde{\mu}_1 - \tilde{\mu}_2)^2 = (w^T \mu_1 - w^T \mu_2)^2 = w^T (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T w$$

Difference of projected means

$$= w^T S_B w.$$

$$\therefore J(w) = \frac{w^T S_B w}{w^T S_w w}$$

between class scatter.
within class scatter

$$\therefore \frac{dJ(w)}{dw} = 0 \Rightarrow \frac{d}{dw} \left[\frac{w^T S_B w}{w^T S_w w} \right].$$

$$= \frac{w^T S_W w \frac{d}{dw} (w^T S_B w) - w^T S_B w \frac{d}{dw} (w^T S_W w)}{(w^T S_B w)^2}$$

$$\Rightarrow (w^T S_W w) (2 S_B w) - (w^T S_B w) (2 S_W w) = 0$$

dividing by $w^T S_W w$

$$= S_B w - \frac{w^T S_B w}{w^T S_W w} S_W w = 0$$

$$= S_B w - \lambda S_W w = 0$$

Multiply by S_W^{-1} .

$$\therefore S_W^{-1} S_B w - \lambda w = 0 \quad \text{value}$$

$$\therefore S_W^{-1} S_B w = \lambda w \quad \text{eigen of } S_W^{-1} S_B \text{ yield.}$$

$$w^+ = \underset{w}{\text{argmax}} \left(\frac{w^T S_B w}{w^T S_W w} \right) = S_W^{-1} (\mu_1 - \mu_2)$$

ex:- LDA ex:- $x_1 = (4, 1), (2, 4), (2, 3), (3, 6), (4, 4)$
 $x_2 = (9, 10), (6, 8), (9, 5), (8, 7), (10, 8)$

1) Calculate mean of each class, therefore projection.

$\mu_1 =$ $\begin{matrix} 4 & 1 \\ 2 & 4 \\ 2 & 3 \\ 3 & 6 \\ 4 & 4 \end{matrix}$	$\mu_2 =$ $\begin{matrix} 9 & 10 \\ 6 & 8 \\ 9 & 5 \\ 8 & 7 \\ 10 & 8 \end{matrix}$
3.60	$8.40, 7.60$

$$S_i = \sum_{x \in w_i} (x - \mu_i)(x - \mu_i)^T$$

$$S_1 = \frac{1}{N} \sum (x_i - \mu_1)(x_i - \mu_1)^T \rightarrow \text{don't forget to average.}$$

* C-classes:-

□ Fisher's LDA generalizes very gracefully for C-class problems.

⇒ Instead of 1 projection y , we seek $(C-1)$ projections $[y_1, y_2, \dots, y_{C-1}]$ by means of $(C-1)$ projection vectors w_i which can be arranged by columns into a projection matrix $W = [w_1 | w_2 | \dots | w_{C-1}]$

$$y_i = w_i^T x \Rightarrow y = W^T x$$

1. Form the data matrix and labels

$$X = \begin{bmatrix} (x^{(1)})^T \\ (x^{(2)})^T \\ \vdots \\ (x^{(n)})^T \end{bmatrix} \in \mathbb{R}^{n \times d}, \quad y \in \{1, \dots, C\}^n$$

2. Compute class counts and means

For each class c :

- $n_c = |\{i : y_i = c\}|$
- $\mu_c = \frac{1}{n_c} \sum_{i: y_i = c} x^{(i)} \in \mathbb{R}^d$

Overall mean:

$$\mu = \frac{1}{n} \sum_{i=1}^n x^{(i)}$$

3. Within-class scatter matrix

$$S_W = \sum_{c=1}^C \sum_{i: y_i = c} (x^{(i)} - \mu_c)(x^{(i)} - \mu_c)^T \in \mathbb{R}^{d \times d}$$

(Optionally divide by $n - C$ to make it a covariance-like estimate—does not affect the eigenvectors.)

4. Between-class scatter matrix

$$S_B = \sum_{c=1}^C n_c (\mu_c - \mu)(\mu_c - \mu)^T \in \mathbb{R}^{d \times d}$$

5. Solve the generalized eigenproblem

Find eigenpairs (λ, w) of

$$S_W^{-1} S_B w = \lambda w$$

(or equivalently solve $S_B w = \lambda S_W w$).

6. Choose projection dimension

Max possible discriminant directions is $\min(d, C - 1)$.

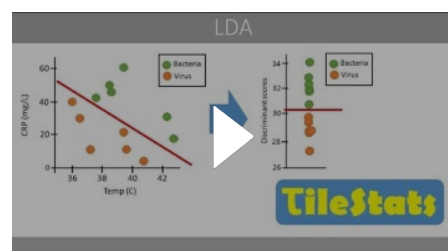
Take the top $k \leq C - 1$ eigenvectors with largest eigenvalues and form

$$W = [w_1, \dots, w_k] \in \mathbb{R}^{d \times k}$$

7. Project the data

$$Z = XW \in \mathbb{R}^{n \times k}$$

How to use LDA as a classifier : [Linear discriminant analysis \(LDA\) - simply explained](#)



⇒ The generalization of within class scatter (variance) is

$$S_w = \sum_{i=1}^C S_i$$

$$\text{where } S_i = \sum_{x \in w_i} (x - \mu_i)(x - \mu_i)^T \text{ \& } \mu_i = \frac{1}{N_i} \sum_{x \in w_i} x_i$$

⇒ Generalization for between class scatter can be found via.

$$S_b = \frac{1}{C} \sum_{i=1}^C (\mu_i - \mu)(\mu_i - \mu)^T$$

⇒ where μ is the mean of all the means.

$$\text{then } J = \frac{w^T S_b w}{w^T S_w w} \quad \left(\begin{array}{l} \text{Usually } S_w \text{ is assumed} \\ \text{to have same covariance} \\ \text{across all classes} \end{array} \right)$$

$$w^* \Rightarrow \boxed{S_w^{-1} S_b w = \lambda w}$$

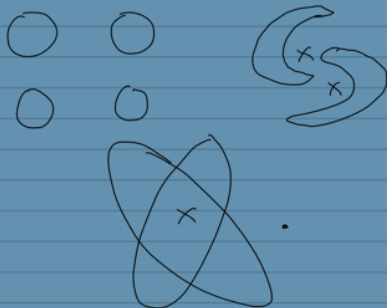
⊠ LDA produces at most $C-1$ feature projections.

∵ S_b is the sum of C matrices of rank one or less

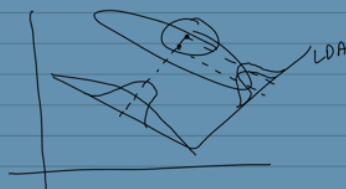
$$\hookrightarrow \therefore \text{rank}(S_b) \leq C-1.$$

⊠ LDA assumes unimodal gaussian likelihood.

∴ LDA can be derived from MLE for the case of normal class conditional densities with equal covariance matrices.



⊠ LDA will fail when the discriminatory feature is not in the mean, but rather in the variance of the data.



This image shows a full page of a notebook or ledger. The background is a solid light blue color. There are horizontal ruling lines spaced evenly across the page. Two vertical red lines run down the left side of the page, creating a narrow margin area. The rest of the page is open space for writing.